

Manikanta Cheruku

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EDUCATION

Sree Dattha Institute of Engineering and Science

Hyderabad, India

B.Tech Computer Science and Business Systems

2023 - 2027

CGPA: 7.0/10 | Self-Study: System Design, RAG Architectures, Vector Databases, LLM Integration

SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI/ML: Ollama, Whisper STT, ChromaDB, SentenceTransformers, LangChain, XGBoost, SHAP, RAG, Vector Embeddings, Agentic AI, Local LLM Inference

Backend: FastAPI, REST API Design, Async Programming, Node.js

Frontend: React.js, Electron, Vite, Tailwind CSS, HashRouter

Databases: PostgreSQL, SQLite, ChromaDB (Vector Database)

Cloud & DevOps: Git, GitHub, Docker, Linux, Render, CI/CD

Methodologies: Agile, OOP, REST API Design, System Design, TDD Basics

PROJECTS

SEVEN [Live]

2025 - Present

Python, FastAPI, Electron, React, Whisper, Ollama, ChromaDB

- Developed a **fully offline AI voice assistant** for Windows integrating **4 AI components** (Whisper STT, Ollama LLM, ChromaDB vector DB, SentenceTransformers) with 100% zero cloud dependency, eliminating recurring API costs and ensuring **full user data privacy**.
- Architected a local **RAG memory system** with ChromaDB enabling semantic search across conversation history. Sub-second vector retrieval on consumer hardware (8GB RAM, no GPU required)
- Implemented **30+ FastAPI endpoints** orchestrating voice pipeline, license validation, telemetry sync, and IPC between Electron shell and Python backend with PostgreSQL on Render
- Shipped solo as complete installable Windows product with under 5 minute setup time product.** Multi-step onboarding wizard, tier-based licensing, referral system. Released publicly on GitHub.

AgentFetch [Live]

Python, FastAPI, React, REST APIs, CLI

- Crafted** agentic data pipeline integrating **2 live APIs** (crypto + weather) with async FastAPI backend, React dashboard, and developer CLI.
- Designed extensible agent interface layer** supporting future LLM orchestration with **zero architectural refactoring** required

Hackathon Team Performance Predictor [Github]

XGBoost, SHAP, Scikit-learn, Pandas

- Constructed** end-to-end ML pipeline achieving **86% classification accuracy** on team performance prediction. XGBoost trained on **3 engineered feature categories**: commit velocity, sleep patterns, communication frequency
- Implemented **SHAP interpretability** producing per-prediction feature attribution scores, making black-box model decisions fully auditable

EXPERIENCE

Solo Developer | SEVEN (Independent Product)

Jan 2026 - Present

Python, FastAPI, Electron, React, Whisper, Ollama, ChromaDB

- Gained expertise in **end-to-end product development expertise** from system design through **deployment and ongoing maintenance**.
- Mastered **AI integration patterns** combining speech recognition, language models, and vector databases into unified production pipelines.

LANGUAGES

English (Fluent) · Hindi (Fluent) · Telugu (Native) · Spanish (Basic)